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### Methods And Systems For Robust, Automated Tuning Of Feedforward Motion Control Parameters

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#### Abstract

Methods and systems for automatically tuning parameters of a feedforward based controller employed by a wafer positioning system in semiconductor processing equipment are described herein. One or more feedforward controller parameter values are selected to provide a desired motion performance at any location in the workspace of the wafer positioning system. The tuned feedforward controller operates in combination with a feedback controller to provide stable control of wafer motion. In one aspect, feedforward controller parameter values are selected to minimize a cost function including a simulated positioning performance associated with a first set of feedforward controller parameter values subtracted from a sum of a simulated positioning performance associated with an updated set of feedforward controller parameter values and a measured positioning performance associated with the first set of feedforward controller parameter values. In some examples, positioning performance is characterized by settling time.

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## **Background/Summary**

CROSS REFERENCE TO RELATED APPLICATION [0001] The present application for patent claims priority under 35 U.S.C. § 119 from U.S. provisional patent application Ser. No. 63/561,746, filed Mar. 6, 2024, the subject matter of which is incorporated herein by reference in its entirety.

### **TECHNICAL FIELD**

[0002] The described embodiments relate to systems for specimen handling, and more particularly to methods and systems for improved specimen positioning performance.

### **BACKGROUND INFORMATION**

[0003] Semiconductor devices such as logic and memory devices are typically fabricated by a sequence of processing steps applied to a specimen. The various features and multiple structural levels of the semiconductor devices are formed by these processing steps. For example, lithography, among others, is one semiconductor fabrication process that involves generating a pattern on a semiconductor wafer. Additional examples of semiconductor fabrication processes include, but are not limited to, chemical-mechanical polishing, etch, deposition, and ion implantation. Multiple semiconductor devices may be fabricated on a single semiconductor wafer and then separated into individual semiconductor devices.

[0004] A lithographic process, as introduced above, is performed to selectively remove portions of a resist material overlaying the surface of a wafer, thereby exposing underlying areas of the specimen on which the resist is formed for selective processing such as etching, material deposition, implantation, and the like. Therefore, in many instances, the performance of the lithography process largely determines the characteristics (e.g., dimensions) of the structures formed on the specimen. Consequently, the trend in lithography is to design systems and components (e.g., resist materials) that are capable of forming patterns having ever smaller dimensions. In particular, the resolution capability of the lithography tools is one primary driver of lithography research and development.

[0005] Measurement processes based on optical radiation, x-ray radiation, or electron based bombardment are used at various steps during a semiconductor manufacturing process to detect defects on wafers to promote higher yield. Optical metrology techniques offer the potential for high throughput without the risk of sample destruction. A number of optical metrology based techniques including scatterometry implementations and associated analysis algorithms to characterize device geometry have been described.

[0006] A wafer is positioned within a wafer processing tool (e.g., lithography tool, etch tool, inspection tool, metrology tool, etc.) by a multiple degree of freedom wafer positioning system. In many embodiments, a wafer positioning system includes two long stroke axes capable of moving the wafer in a plane approximately parallel with the top surface of the wafer. The two long stroke axes are able to move the wafer in the plane of such that any location on the wafer surface is addressable by a wafer processing subsystem, e.g., a metrology subsystem, an inspection subsystem, a lithography subsystem, an etch subsystem, a deposition subsystem, etc.

[0007] Typically, the position of the wafer within a wafer processing system is maintained by an active controller that calculates control signals based on feedback, e.g., position feedback, velocity feedback, etc. Feedback control signals are driven by errors calculated as the difference between measured motion variables and desired values of those motion variables, e.g., difference between

the measured position of the wafer and the desired position of the wafer. Feedback control enables tracking of motion command signals, rejection of disturbance forces, and stable interaction between forces generated by multiple axes of a multiple degree of freedom wafer positioning system.

[0008] Unfortunately, the performance of a wafer positioning system controlled based on feedback signals alone has some limitations. A feedback control scheme is by definition reactive in nature because the controller is driven by errors. The errors must first be measured and then compensated by applying corrective force to the motion system. These actions take time, and thus limit tracking performance. Furthermore, the ability of a feedback control system to react quickly is limited by a number of practical physical considerations. For example, the finite stiffness of mechanical structures limits the achievable bandwidth of a stable and robust feedback controller. This, in turn, limits how quickly and effectively, a feedback based controller is able to track motion commands. Although, additional control elements such as input shaping and notch filters help to reduce the sensitivity of feedback controllers to structural dynamics, performance gains are limited and often come at a cost of decreased robustness.

[0009] In some motion control systems, feedforward control is implemented, in addition to feedback control, to improve motion performance beyond the limitations imposed by feedback based control. A feedforward based control element calculates control signals based on knowledge of the desired motion trajectory, e.g., velocity feedforward, acceleration feedforward, jerk feedforward, etc. Feedforward control signals are driven by anticipated forces required to achieve a desired motion trajectory. Feedforward control is by nature proactive because feedforward control signals are driven by the desired motion trajectory, which is available instantaneously. Thus, control action derived from feedforward based signals is implemented instantaneously on the time scale of physical stage motion.

[0010] Feedforward control has the potential to improve tracking of motion command signals, minimize interaction forces between multiple axes of a multiple degree of freedom wafer positioning system, etc. However, to effectively realize the benefits of feedforward based control, precise system knowledge is required.

[0011] Traditionally, manual tuning of feedforward parameters is achieved based on precise system identification and trial-and-error techniques. Very often, the feedforward control parameters determined for one instance of a wafer positioning system are not effective when applied to another instance of the same wafer positioning system. Mechanical variations among different instances of a wafer positioning system are often large enough that a single set of feedforward control parameters will not achieve a desired motion performance across all instances of the wafer positioning system. As a consequence, each instance of a wafer positioning system must be individually tuned to guarantee acceptable motion performance.

[0012] In addition, feedforward control parameters must be selected such that the desired motion performance is achieved at all positions within the workspace of the multiple degree of freedom wafer positioning system. This compounds the tuning effort. First, system identification must be performed throughout the workspace. Second, a set of feedforward control parameters must be manually tuned to meet the desired motion performance requirements at all positions within the workspace. Often, the set of feedforward control parameters determined manually are not robust to variations in system behavior.

[0013] In a production setting, the effort required to manually tune feedforward control parameters of wafer positioning systems is limiting. Significant amounts of time are required to tune each system, the results are not as robust as desired, and extensive expertise and knowledge of both the system and controller tuning process is required. Improved methods and systems for tuning feedforward based controllers employed by wafer positioning systems in semiconductor processing equipment are desired.

## SUMMARY

[0014] Methods and systems for automatically tuning parameters of a feedforward based controller

employed by a wafer positioning system in semiconductor processing equipment are described herein. Parameter values of a feedforward controller are selected to provide a desired motion performance at any location in the workspace of the wafer positioning system. The tuned feedforward controller operates in combination with a feedback controller to provide stable control of wafer motion in semiconductor processing equipment.

[0015] In some embodiments, a motion controller includes both a feedback controller and a feedforward controller. Typically, the feedback control element operates on motion errors, i.e., the difference between a desired motion and the actual, measured motion. The feedback control element generates motion commands based on the motion errors. In contrast, a feedforward control element generates motion commands based on the desired motion, directly. The motion commands generated by the feedback and feedforward control elements are communicated to a positioning subsystem. In response, forces/torques are generated by actuators of the wafer positioning system in response to the motion commands.

[0016] In some embodiments, a feedforward controller generates control signals from one or more motion parameters, e.g., position, velocity, acceleration, jerk, etc. In general, a feedforward controller may be configured to generate motion control signals from any number of motion parameters, including higher order derivatives of position.

[0017] In one aspect, parameters of a feedforward based controller employed by a wafer positioning system are automatically tuned. Values of parameters of a feedforward based controller are selected to minimize a cost function including a simulated positioning performance during a predetermined movement based on a first set of values of the feedforward control parameters subtracted from a sum of a simulated positioning performance during the predetermined movement based on an updated set of values of the feedforward control parameters and a measured positioning performance during the predetermined movement based on the first set of values of the feedforward control parameters. The number of feedforward parameters undergoing optimization can be one or more.

[0018] The inventors have discovered that the difference between simulated positioning performance at two different sets of values of feedforward control parameters and the difference between actual positioning performance at the same two different sets of values of the feedforward control parameters match very closely. Thus, the simulated positioning performance at both initial and updated sets of values of feedforward control parameters and the actual positioning performance at the initial set of values accurately estimates the actual positioning performance at the updated set of values. Hence, a minimization of the aforementioned cost function is performed to arrive at an updated set of values of feedforward parameters that achieves a desired motion performance.

[0019] In some examples, positioning performance is characterized by settling time, i.e., the time required for the position of a wafer to settle to the desired position within specification after the nominal movement is complete.

[0020] In some embodiments, the feedforward control parameters undergoing optimization are associated with a single input, single output controller configured to control one degree of freedom of a wafer positioning system.

[0021] In some embodiments, the feedforward control parameters undergoing optimization are associated with a multi-input, multi-output controller configured to control two or more degrees of freedom of a wafer positioning system.

[0022] In a further aspect, optimization of feedforward control parameters is associated with actual motion performance and simulated motion performance at multiple points in the workspace of wafer positioning system. In this manner, the optimized feedforward control parameter values ensure desired motion performance at different points in the workspace of the wafer positioning system.

[0023] The foregoing is a summary and thus contains, by necessity, simplifications, generalizations

and omissions of detail. Consequently, those skilled in the art will appreciate that the summary is illustrative only and is not limiting in any way. Other aspects, inventive features, and advantages of the devices and/or processes described herein will become apparent in the non-limiting detailed description set forth herein.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0024] FIG. 1 is a diagram illustrative of a measurement system including a wafer positioning system configured to automatically tune feedforward controller parameter values associated with a feedforward controller employed to control motion of the wafer positioning system as described herein.

[0025] FIG. 2 is a diagram illustrative of a control scheme for controlling the motion of a wafer positioning system of a semiconductor measurement system in one embodiment.

[0026] FIG. 3 is a diagram illustrative of a feedforward controller for generating feedforward command signals in one embodiment.

[0027] FIG. 4A is a plot illustrative of an exemplary desired position trajectory associated with a displacement of approximately 120 millimeters.

[0028] FIG. 4B is a plot illustrative of an exemplary desired velocity trajectory associated with the desired position trajectory depicted in FIG. 4A.

[0029] FIG. 4C is a plot illustrative of an exemplary desired acceleration trajectory associated with the desired position trajectory depicted in FIG. 4A.

[0030] FIG. 4D is a plot illustrative of an exemplary desired jerk trajectory associated with the desired position trajectory depicted in FIG. 4A.

[0031] FIG. 5 is a simplified diagram of a top view of a workspace of a wafer positioning system.

[0032] FIG. 6 is a plot illustrative of position error as a function of time during a move and settle wafer stage movement from a location in the workspace of a wafer positioning system.

[0033] FIG. 7 illustrates a flowchart of an exemplary method useful for automatically tuning feedforward motion control parameter values in at least one novel aspect.

### DETAILED DESCRIPTION

[0034] Reference will now be made in detail to background examples and some embodiments of the invention, examples of which are illustrated in the accompanying drawings.

[0035] Methods and systems for automatically tuning parameters of a feedforward based controller employed by a wafer positioning system in semiconductor processing equipment are described herein. Parameter values of a feedforward controller are selected to provide a desired motion performance at any location in the workspace of the wafer positioning system. The tuned feedforward controller operates in combination with a feedback controller to provide stable control of wafer position in semiconductor processing equipment.

[0036] FIG. 1 is a diagram illustrative of a measurement system **100** including a machine frame **101**, a wafer positioning system **110**, a computing system **190**, and a semiconductor measurement subsystem **160**. As depicted in FIG. 1, wafer positioning system **110** is mechanically coupled to machine frame **101**. Wafer positioning system **110** includes multiple motion stages that operate in coordination to move wafer **119** with respect to machine frame **101** in six degrees of freedom. Although wafer positioning system **110** is described herein as a six degree of freedom positioning system, in general, any wafer positioning system employed to locate a wafer in at least one degree of freedom is contemplated within the scope of this patent document.

[0037] As depicted in FIG. 1, wafer positioning system **110** includes a long stroke base stage including base reference structures **111A** and **111B** and X-frame **112**. Base reference structures **111A** and **111B** are mechanically coupled to machine frame **110**. X-frame **112** is mechanically

constrained by bearing elements (not shown) to move with respect to base reference structures **111A** and **111B** in one degree of freedom that is approximately aligned with the X-direction depicted in FIG. 1. A base stage drive mechanism (not shown) generates drive forces to move X-frame **112** with respect to base reference structures **111A** and **111B**.

[0038] Wafer positioning system **110** also includes a long stroke intermediate stage including Y-frame **114** moveable with respect to X-frame **112**. Y-frame **114** is mechanically constrained by bearing elements (not shown) to move with respect to X-frame **112** in one degree of freedom that is approximately aligned with the Y-direction depicted in FIG. 1. An intermediate stage drive mechanism (not shown) generates drive forces to move Y-frame **114** with respect to X-frame **112**.

[0039] By way of non-limiting example, bearing elements of the base and intermediate stages may include mechanical linear bearings, linear air bearings, linear magnetic bearings, etc. In general, any suitable linear bearing arrangement may be contemplated within the scope of this patent document.

[0040] By way of non-limiting example, base stage and intermediate stage drive mechanisms may include a linear motor, a rotary motor and ball spindle, a rotary motor and belt drive, etc. In general, any suitable stage drive mechanism may be contemplated within the scope of this patent document. The base stage and intermediate stage are long stroke motion stages, e.g., total stroke of more than 100 millimeters.

[0041] Wafer positioning system **110** also includes a tip/tilt/Z stage including tip/tilt/Z stage **117** moveable with respect to Y-frame **114**. Tip/tilt/Z stage **117** includes three linear actuators **117A-C** configured to independently move tip/tilt/Z stage **117** linearly with respect to intermediate stage **114** in the Z-direction and rotate tip/tilt/stage **117** about the X and Y axes. Tip/tilt/Z stage **117** is a short stroke motion stage, e.g., total stroke of actuators **117A-C** is less than 10 millimeters. By way of non-limiting example, linear actuators **117A-C** may include a piezoelectric linear motor, a Lorentz coil motor, etc. In general, any suitable stage drive mechanism may be contemplated within the scope of this patent document.

[0042] Wafer positioning system **110** also includes a rotary stage including wafer chuck **120** constrained to rotate with respect to tip/tilt/Z stage **117**. A rotary bearing (not shown) is configured to constrain the movement of wafer chuck **120** with respect to tip/tilt/Z stage **117** to rotation about the Z-axis. By way of non-limiting example, bearing elements of the rotary stage may include mechanical bearings, air bearings, magnetic bearings, etc. In general, any suitable rotary bearing arrangement may be contemplated within the scope of this patent document.

[0043] A rotary motor assembly **118** is configured to provide rotational torque to rotate wafer chuck **120** with respect to tip/tilt/Z stage **117** about the Z-axis. By way of non-limiting example, rotary motor assembly **118** may include a rotary motor and belt drive arrangement, a direct drive electric motor having rotor and stator elements mounted to the wafer chuck **120** and tip/tilt/Z stage **117**, respectively, or vice-versa, etc. In general, any suitable rotary drive arrangement may be contemplated within the scope of this patent document.

[0044] Wafer **119** is clamped on the top surface of wafer chuck **120**. In this manner, wafer positioning system **110** is configured to move wafer **119** in six degrees of freedom: linear motion aligned with the X, Y, and Z axes, which are orthogonal to one another, and rotational motion about the X, Y, and Z axes.

[0045] As depicted in FIG. 1, wafer positioning system **110** includes a dynamic cable system including dynamic cable **115** coupled to machine frame **101** and X-frame **112** and dynamic cable **116** coupled to X-frame **112** and Y-frame **114**. Dynamic cable **115** provides routing for electrical wiring, positively pressurized air conduits, negatively pressurized air conduits, etc., between machine frame **101** and X-frame **112**. Dynamic cable **116** provides routing for electrical wiring, positively pressurized air conduits, negatively pressurized air conduits, etc., between X-frame **112** and Y-frame **114**. In the embodiment depicted in FIG. 1, negatively pressurized air **123** is routed from machine frame **101** through dynamic cable **115**, through X-frame **112**, through dynamic cable

**116**, through Y-frame **114**, through vacuum feedthrough **122** to wafer chuck **120**. In this manner, vacuum is provided from machine frame **101** to wafer chuck **120** to maintain wafer **119** clamped onto the top surface of wafer chuck **120**.

[0046] Computing system **190** receives signals **195** indicative of motion of wafer **119** with respect to machine frame **101**, e.g., position, velocity, acceleration, or any combination thereof. Based on signals **195** computing system **190** determines control command signals **196** communicated to one or more stages of wafer positioning system **110**. In response, wafer positioning system **110** generates drive forces/torques that cause motion of wafer **119** relative to machine frame **101**. In this manner, computing system **190** controls the motion of wafer **119** with respect to machine frame **101**.

[0047] FIG. 2 is a diagram illustrative of a control scheme **130** for controlling the motion of a wafer positioning system of a semiconductor measurement system in one embodiment. Control scheme **130** includes a controller **137**, a motion subsystem **133**, and motion sensors **134**. In one example, controller **137** is implemented by computing system **190** depicted in FIG. 1. In this example, motion subsystem **133** is implemented by the drive and mechanical elements of wafer positioning system **110** depicted in FIG. 1, e.g., electrical drive components, such as amplifiers and motors, and mechanical components such as frames, bearings, cabling, etc. Also in this example motion sensors **134** are implemented by the motion sensors embedded in wafer positioning system **110**, e.g., encoders, interferometers, inductive probes, capacitive probes, etc., employed to measure movements of wafer chuck **120** with respect to machine frame **101**.

[0048] As depicted in FIG. 2, controller **137** includes both a feedback control element **132** and a feedforward control element **131**. In one example, computing system **190** determines a motion error signal **136** as the difference between a desired motion command signal **135** and motion signals **195** indicative of the current state of motion of wafer **119** with respect to machine frame **101**. The motion error signal **136** is provided as input to feedback controller **132**. Computing system **190** implementing feedback controller **132** computes feedback command signals **138** based on the motion error signal **136**. In addition, desired motion command signal **135** is provided as input to feedforward controller **131**. Computing system **190** implementing feedforward controller **131** computes feedforward command signals **139** based on the desired motion command signal **135**. Computing system **190** determines control command signals **196** as the sum of feedback command signals **138** and feedforward command signals **139**. The control command signals **196** are communicated to wafer positioning subsystem **110**. In response, forces/torques are generated by actuators of wafer positioning subsystem **110** that cause motion of wafer **119** with respect to machine frame **101**. The motion is measured by motion sensors **134**. Motion signals **195** indicative of the current state of motion of wafer **119** with respect to machine frame **101** are communicated to computing system **190** for another iteration of the control loop.

[0049] FIG. 3 is a diagram illustrative of a feedforward controller **131** for generating feedforward command signals **139** in one embodiment. As depicted in FIG. 3, feedforward controller **131** generates control signals from several motion parameters including position, velocity, acceleration, and jerk. In general, feedforward controller **131** may be configured to generate motion control signals from any number of motion parameters, including higher order derivatives of position.

[0050] As depicted in FIG. 3, desired position trajectory **135** is communicated to first order differentiation block **142**. In one example, computing system **190** determines a first order derivative of desired position trajectory **135** with respect to time to generate a desired velocity trajectory **153**. Similarly, desired position trajectory **135** is communicated to second order differentiation block **143**. In one example, computing system **190** determines a second order derivative of desired position trajectory **135** with respect to time to generate a desired acceleration trajectory **154**. Similarly, desired position trajectory **135** is communicated to third order differentiation block **144**. In one example, computing system **190** determines a third order derivative of desired position trajectory **135** with respect to time to generate a desired jerk

trajectory **155**.

[0051] FIG. **4A** is a plot illustrative of an exemplary desired position trajectory **135** for a displacement of approximately 120 millimeters. FIG. **4B** is a plot illustrative of an exemplary desired velocity trajectory **153** associated with desired position trajectory **135**. FIG. **4C** is a plot illustrative of an exemplary desired acceleration trajectory **154** associated with desired position trajectory **135**. FIG. **4D** is a plot illustrative of an exemplary desired jerk trajectory **155** associated with desired position trajectory **135**.

[0052] As depicted in FIG. **3**, desired motion command signal **135** includes a desired position trajectory for a displacement of wafer **119** from one position to another. Desired motion command signal **135** is provided directly to position feedforward control element **145**. Position feedforward control element **145** includes a position feedforward control parameter,  $K_{sub.P}$ . In one example, computing system **190** determines position feedforward command signal **149** as the desired position command signal **135** multiplied by position feedforward control parameter,  $K_{sub.P}$ .

[0053] Similarly, desired velocity trajectory **153** is provided directly to velocity feedforward control element **146**. Velocity feedforward control element **146** includes a velocity feedforward control parameter,  $K_{sub.V}$ . In one example, computing system **190** determines velocity feedforward command signal **150** as the desired velocity trajectory **153** multiplied by velocity feedforward control parameter,  $K_{sub.V}$ .

[0054] Similarly, desired acceleration trajectory **154** is provided directly to acceleration feedforward control element **147**. Acceleration feedforward control element **147** includes an acceleration feedforward control parameter,  $K_{sub.A}$ . In one example, computing system **190** determines acceleration feedforward command signal **151** as the desired acceleration trajectory **154** multiplied by acceleration feedforward control parameter,  $K_{sub.A}$ .

[0055] Similarly, desired jerk trajectory **155** is provided directly to jerk feedforward control element **148**. Jerk feedforward control element **148** includes a jerk feedforward control parameter,  $K_{sub.J}$ . In one example, computing system **190** determines jerk feedforward command signal **152** as the desired jerk trajectory **155** multiplied by jerk feedforward control parameter,  $K_{sub.J}$ .

[0056] In one aspect, parameters of a feedforward based controller employed by a wafer positioning system are automatically tuned. Parameter values of a feedforward controller are selected to provide a desired motion performance at any location in the workspace of the wafer positioning system. The tuned feedforward controller operates in combination with a feedback controller to provide stable control of wafer position in semiconductor processing equipment.

[0057] Values of parameters of a feedforward based controller, e.g.,  $K_{sub.p}$ ,  $K_{sub.v}$ ,  $K_{sub.a}$ , and  $K_{sub.J}$  depicted in FIG. **3**, are selected to minimize a cost function including a simulated positioning performance during a predetermined movement based on a first set of values of the feedforward control parameters subtracted from a sum of a simulated positioning performance during the predetermined movement based on an updated set of values of the feedforward control parameters and a measured positioning performance during the predetermined movement based on the first set of values of the feedforward control parameters. Equation (1) illustrates the cost function to be minimized, where  $\sup S Y(x)$  is the simulated positioning performance based on the updated set of values of the feedforward control parameters,  $\sup SY(x_{sub.0})$  is the simulated positioning performance based on the first set of values of the feedforward control parameters, and  $\sup MY(x_{sub.0})$  is the measured positioning performance based on the first set of values of the feedforward control parameters.

$$[00001] \min .\text{Math.} \quad \sup Y(x) - \sup Y(x_0) + \sup M Y(x_0) .\text{Math.} \quad (1)$$

[0058] Experience has shown that it is extremely difficult to build a simulation model of positioning performance that matches actual measured positioning performance when high positioning performance is required. Thus, it is difficult to use a simulated model of positioning performance to directly predict values of feedforward control parameters that deliver high



positioning performance in actual practice.

[0059] However, the inventors have discovered that the difference between simulated positioning performance at two different sets of values of feedforward control parameters and the difference between actual positioning performance at the same two different sets of values of the feedforward control parameters match very closely as illustrated by the relationship of Equation (2).

$$[00002] \quad \text{Math. } {}^M Y(x) - {}^M Y(x_0) \approx \text{Math. } {}^S Y(x) - {}^S Y(x_0) \quad (2)$$

[0060] In some examples, positioning performance is characterized by settling time, i.e., the time required for the position of wafer **119** to settle to the desired position within specification after the nominal movement is complete. In one example,  $Y$ , is the settling time. It is desirable to select an updated set of feedforward parameter values,  $x$ , that minimize the actual settling time evaluated at those parameter values as illustrated by Equation (3).

$$[00003] \quad \min \text{Math. } {}^M Y(x) \quad (3)$$

[0061] Based on the observation described with reference to Equation (2), the minimization described by Equation (3) is recast as the minimization illustrated by Equation (1). In some examples, the desired performance is achieved in one step. In some other examples, the minimization is performed iteratively to arrive at an updated set of values of the feedforward parameters that achieves the desired motion performance.

[0062] As described hereinbefore, the number of feedforward parameters undergoing optimization can be one or more. In some examples, a feedforward controller is characterized by a single feedforward parameter, e.g., acceleration feedforward,  $K_{sub.A}$ . In these examples, only the single feedforward parameter value is optimized. In other examples, a feedforward controller is characterized by more than one feedforward parameter, e.g., feedforward controller **131** depicted in FIG. **3**. In these examples, multiple feedforward parameter values are optimized.

[0063] In some embodiments, the feedforward control parameters undergoing optimization is associated with a single input, single output controller configured to control one degree of freedom of a single degree of freedom or multi-degree of freedom wafer positioning system. In the embodiment described with reference to FIG. **1**, wafer positioning system **110** is a six degree of freedom positioning system. In this embodiment, six different stage motions are coordinated to move wafer **119** arbitrarily in the six degree of freedom workspace of the wafer positioning system. In some embodiments, each of the six different stage motions are independently controlled by corresponding single input, single output controllers. In one example, motion controller **137** is characterized by a set of feedback and feedforward parameter values employed to control the movements of the X-stage of wafer positioning system **110**. In this same example, another instance of motion controller **137** is characterized by a different set of feedback and feedforward parameter values employed to control the movements of the Y-stage of wafer positioning system **110**.

[0064] In some embodiments, the feedforward control parameters undergoing optimization is associated with a multi-input, multi-output controller configured to control two or more degrees of freedom of a multi-degree of freedom specimen positioning system. In one example, motion controller **137** is a multi-input, multi-output controller characterized by a set of feedback and feedforward parameter values employed to control the movements of the X-stage and Y-stage of wafer positioning system **110** in combination.

[0065] In a further aspect, optimization of feedforward control parameters is associated with actual motion performance and simulated motion performance at multiple points in the workspace of a specimen positioning system. In this manner, the optimized feedforward control parameter values ensure desired motion performance at different points in the workspace of the specimen positioning system. Thus, the optimized feedforward control parameter values ensure robust motion control over the entire workspace of the specimen positioning system.

[0066] FIG. **5** is a top view of the workspace **120** of a specimen positioning system, e.g., wafer positioning system **110**, in the X and Y directions. The dots represent sampling points within the

workspace envelope **122**. For example, dot **121** is the middle of the workspace. In the example movement, wafer **119** is translated in Y-direction.

[0067] In some examples, the minimization illustrated in FIG. **1** is performed on a data set that includes multiple sampling points in the workspace of a specimen positioning system, e.g., the sampling points depicted in FIG. **5**. In some of these examples, a step and settle movement, e.g., the movement described with reference to FIG. **4A**, is initiated from each sampling point within the workspace envelope **122**.

[0068] In some examples, the motion performance metric is settling time. FIG. **6** is a plot **180** illustrative of position error as a function of time during a move and settle movement in the Y-direction from a single sample point in the workspace of wafer positioning system **110**. Plotline **181** illustrates position error during the move and settle movement before optimization of feedforward parameter values. Plotline **182** illustrates the position error during the move and settle movement after optimization of feedforward parameter values. As illustrated in FIG. **6**, the settling time is the time elapsed from the moment the nominal movement, i.e., the commanded movement, is terminated, to the moment the position error settles within the required position error specification. In some other embodiments, the settling time is the time elapsed from the moment the nominal movement, i.e., the commanded movement, is initiated to the moment the position error settles within the required position error specification. The settling time,  $T_{sub.s}$ , depicted in FIG. **6** is associated with the pre-determined movement before optimization of feedforward parameter values. As depicted in FIG. **6**, the magnitude of the position error is much larger before optimization of feedforward parameter values. In addition, the settling time is much larger before optimization of feedforward parameter values. In general, the smaller the position error specification, the optimization of feedforward parameter values becomes more critical. In some embodiments, the position error specification is 50 nanometers or less. In these embodiments, the difference between settling time before and after optimization of feedforward parameter values is dramatic.

[0069] FIG. **1** illustrates a simplified schematic of an optical metrology or inspection system **160** positioned to inspect wafer **119** or perform measurements of structures formed on wafer **119**. In some embodiments, system **160** is configured as a scanning system. In some other embodiments system **160** is configured as a point to point measurement system. In the depicted embodiment, wafer **119** is vacuum clamped to wafer chuck **120**, while wafer chuck **120** is moved in multiple degrees of freedom by wafer positioning system **110**.

[0070] As illustrated in FIG. **1**, wafer **119** is illuminated by a normal incidence beam **163** generated by one or more illumination sources **161**. Alternatively, the illumination subsystem may be configured to direct the beam of light to the specimen at an oblique angle of incidence. In some embodiments, system **160** may be configured to direct multiple beams of light to the specimen such as an oblique incidence beam of light and a normal incidence beam of light. The multiple beams of light may be directed to the specimen substantially simultaneously or sequentially.

[0071] Illumination source **161** may include, by way of example, a laser, a diode laser, a helium neon laser, an argon laser, a solid state laser, a diode pumped solid state (DPSS) laser, a xenon arc lamp, a gas discharging lamp, and LED array, or an incandescent lamp. The light source may be configured to emit near monochromatic light or broadband light. The illumination subsystem may also include one or more spectral filters that may limit the wavelength of the light directed to the specimen. The one or more spectral filters may be bandpass filters and/or edge filters and/or notch filters.

[0072] Normal incidence beam **163** is focused onto the substrate **110** by an objective lens **164**. System **160** includes collection optics **162** to collect the light scattered and/or reflected by wafer **119** in response to the illumination light **163**. Collection optics **162** focus the collected light onto a detector **165**. The output signals **166** generated by detector **165** are supplied to a computing system **190** for processing the signals and determining the measurement parameter values **197** (e.g.,

material or structural properties, dimensions, presence of particles, etc.). System **160** is presented herein by way of non-limiting example, as wafer positioning system **110** may be implemented within many different electron based, x-ray based, or optical based metrology and inspection systems.

[0073] As illustrated in FIG. **1**, the system **160** is configured as an inspection system or a metrology system. In this manner, the system may be configured to inspect or measure wafers and reticles used as part of a semiconductor manufacturing process. The methods and systems described herein are not limited to the inspection or measurement of semiconductor wafers or reticles, and may be applied to the inspection of other substrates that require accurate, high performance position control for processing.

[0074] As described hereinbefore, automatic tuning of feedforward control parameters is described with reference to motion control of wafer **119**. However, in general, it is contemplated within the scope of this patent document that the methods and systems for automatic tuning of feedforward control parameters described herein may be applied to motion control of other specimens, including, but not limited to, semiconductor reticles, optical elements of a measurement system, such as lenses, detectors, polarizers, etc.

[0075] FIG. **7** illustrates a flowchart of an exemplary method **200** useful for automatically tuning feedforward control parameter values in several novel aspects. By way of non-limiting example, method **200** is described with reference to the measurement system **100** illustrated in FIG. **1** for explanatory purposes. Although, the description of system **100** includes references to specific hardware elements employed to achieve the elements of method **200**, many other hardware elements known to persons of ordinary skill in the art may be contemplated to achieve an analogous result. Hence, any of the referenced hardware elements presented herein may be substituted, consolidated, modified, or eliminated without exceeding the scope of the description provided herein. Similarly, some of the elements of method **200** and the order of presentation of the elements of method **200** relate to the use of specific hardware elements described with reference to system **100**. However, as many other hardware elements known to persons of ordinary skill in the art may be contemplated to achieve an analogous result, some of the method elements and the order of presentation of the method elements may be substituted, consolidated, modified, or eliminated without exceeding the scope of the description provided herein.

[0076] In block **201**, a first positioning performance of a specimen positioning system is measured during a predetermined movement at each of one or more locations in a workspace of the specimen positioning system. The specimen positioning system including a motion controller controlling the predetermined movement at each of the one or more locations in the workspace. The motion controller includes at least one feedforward control element characterized by one or more feedforward control parameters. The predetermined movement undertaken at a first set of values corresponding to the one or more feedforward control parameters.

[0077] In block **202**, a second positioning performance of the specimen positioning system is simulated during the predetermined movement at each of one or more locations in the workspace based on the first set of values corresponding to the one or more feedforward control parameters.

[0078] In block **203**, a third positioning performance of the specimen positioning system is simulated during the predetermined movement at each of one or more locations in the workspace based on an updated set of values corresponding to the one or more feedforward control parameters.

[0079] In block **204**, the updated set of values corresponding to the one or more feedforward control parameters is selected to minimize a cost function including the second positioning performance subtracted from a sum of the first and third positioning performances.

[0080] In a further embodiment, system **100** includes one or more computing systems **190** employed to perform measurements of actual device structures positioned in accordance with the methods described herein. The one or more computing systems **190** may be communicatively

coupled to the wafer positioning system **110**. In one aspect, the one or more computing systems **190** are configured to receive motion data associated with measurements of the motion of the specimen under measurement.

[0081] It should be recognized that one or more steps described throughout the present disclosure may be carried out by a single computer system **190** or, alternatively, a multiple computer system **190**. Moreover, different subsystems of system **100** may include a computer system suitable for carrying out at least a portion of the steps described herein. Therefore, the aforementioned description should not be interpreted as a limitation on the present invention but merely an illustration.

[0082] In addition, the computer system **190** may be communicatively coupled to the wafer positioning system **110** in any manner known in the art. For example, the one or more computing systems **190** may be coupled to computing systems associated with the wafer positioning system. In another example, the wafer positioning system may be controlled directly by a single computer system coupled to computer system **190**.

[0083] The computer system **190** of measurement system **100** may be configured to receive and/or acquire data or information from the subsystems of the system (e.g., detectors, wafer positioning system, and the like) by a transmission medium that may include wireline and/or wireless portions. In this manner, the transmission medium may serve as a data link between the computer system **190** and other subsystems of system **100** (e.g., memory on-board system **100**, external memory, or other external systems). In this manner, the transmission medium may serve as a data link between the computer system **190** and other systems. For example, the computing system **190** may be configured to receive measurement data from a storage medium (i.e., memory **192** or an external memory) via a data link. For instance, motion measurement results obtained using the sensors of wafer positioning system **110** described herein may be stored in a permanent or semi-permanent memory device (e.g., memory **192** or an external memory). In this regard, the measurement results may be imported from on-board memory or from an external memory system. Moreover, the computer system **190** may send data to other systems via a transmission medium. For instance, control commands **196** or an estimated parameter value **197** determined by computer system **190** may be communicated and stored in an external memory. In this regard, measurement results may be exported to another system.

[0084] Computing system **190** may include, but is not limited to, a personal computer system, mainframe computer system, workstation, cloud based computing system, image computer, parallel processor, or any other device known in the art. In general, the term “computing system” may be broadly defined to encompass any device having one or more processors, which execute instructions from a memory medium.

[0085] Program instructions **194** implementing methods such as those described herein may be transmitted over a transmission medium such as a wire, cable, or wireless transmission link. For example, as illustrated in FIG. 1, program instructions **194** stored in memory **192** are transmitted to processor **191** over bus **193**. Program instructions **194** are stored in a computer readable medium (e.g., memory **192**). Exemplary computer-readable media include read-only memory, a random access memory, solid-state memory, a magnetic or optical memory, or a magnetic tape.

[0086] Various embodiments are described herein for a semiconductor processing system (e.g., an inspection system, a metrology system, a lithographic system, an etch system, etc.). The term “substrate” is used herein to refer to a wafer, a reticle, or any other sample that may be processed (e.g., printed or inspected for defects) by means known in the art.

[0087] As used herein, the term “wafer” generally refers to substrates formed of a semiconductor or non-semiconductor material. Examples include, but are not limited to, monocrystalline silicon, gallium arsenide, and indium phosphide. Such substrates may be commonly found and/or processed in semiconductor fabrication facilities. In some cases, a wafer may include only the substrate (i.e., bare wafer). Alternatively, a wafer may include one or more layers of different

materials formed upon a substrate. One or more layers formed on a wafer may be “patterned” or “unpatterned.” For example, a wafer may include a plurality of dies having repeatable pattern features.

[0088] A “reticle” may be a reticle at any stage of a reticle fabrication process, or a completed reticle that may or may not be released for use in a semiconductor fabrication facility. A reticle, or a “mask,” is generally defined as a substantially transparent substrate having substantially opaque regions formed thereon and configured in a pattern. The substrate may include, for example, a glass material such as quartz. A reticle may be disposed above a resist-covered wafer during an exposure step of a lithography process such that the pattern on the reticle may be transferred to the resist.

[0089] One or more layers formed on a wafer may be patterned or unpatterned. For example, a wafer may include a plurality of dies, each having repeatable pattern features. Formation and processing of such layers of material may ultimately result in completed devices. Many different types of devices may be formed on a wafer, and the term wafer as used herein is intended to encompass a wafer on which any type of device known in the art is being fabricated.

[0090] In one or more exemplary embodiments, the functions described may be implemented in hardware, software, firmware, or any combination thereof. If implemented in software, the functions may be stored on or transmitted over as one or more instructions or code on a computer-readable medium. Computer-readable media includes both computer storage media and communication media including any medium that facilitates transfer of a computer program from one place to another. A storage media may be any available media that can be accessed by a general purpose or special purpose computer. By way of example, and not limitation, such computer-readable media can comprise RAM, ROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium that can be used to carry or store desired program code means in the form of instructions or data structures and that can be accessed by a general-purpose or special-purpose computer, or a general-purpose or special-purpose processor. Also, any connection is properly termed a computer-readable medium. For example, if the software is transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave, then the coaxial cable, fiber optic cable, twisted pair, DSL, or wireless technologies such as infrared, radio, and microwave are included in the definition of medium. Disk and disc, as used herein, includes compact disc (CD), laser disc, optical disc, digital versatile disc (DVD), floppy disk and blu-ray disc where disks usually reproduce data magnetically, while discs reproduce data optically with lasers. Combinations of the above should also be included within the scope of computer-readable media.

[0091] Although certain specific embodiments are described above for instructional purposes, the teachings of this patent document have general applicability and are not limited to the specific embodiments described above. Accordingly, various modifications, adaptations, and combinations of various features of the described embodiments can be practiced without departing from the scope of the invention as set forth in the claims.

## Claims

1. A method comprising: measuring a first positioning performance of a specimen positioning system during a predetermined movement at each of one or more locations in a workspace of the specimen positioning system, the specimen positioning system including a motion controller controlling the predetermined movement at each of the one or more locations in the workspace, the motion controller including at least one feedforward control element characterized by one or more feedforward control parameters, the predetermined movement undertaken at a first set of values corresponding to the one or more feedforward control parameters; simulating a second positioning performance of the specimen positioning system during the predetermined movement at each of

one or more locations in the workspace based on the first set of values corresponding to the one or more feedforward control parameters; simulating a third positioning performance of the specimen positioning system during the predetermined movement at each of one or more locations in the workspace based on an updated set of values corresponding to the one or more feedforward control parameters; and selecting the updated set of values corresponding to the one or more feedforward control parameters to minimize a cost function including the second positioning performance subtracted from a sum of the first and third positioning performances.

**2.** The method of claim 1, wherein the first, second and third positioning performances are positioning errors after a predetermined time after initiation of the predetermined movement.

**3.** The method of claim 1, wherein the first, second, and third positioning performances are positioning errors after a predetermined time after termination of the predetermined movement.

**4.** The method of claim 1, wherein the one or more locations in the workspace includes multiple, different locations in the workspace in a direction aligned with a first degree of freedom of the specimen positioning system.

**5.** The method of claim 4, wherein the one or more locations in the workspace includes multiple, different locations in the workspace in a direction perpendicular with the first degree of freedom.

**6.** The method of claim 1, wherein the workspace of the specimen positioning system includes at least two degrees of freedom.

**7.** The method of claim 1, further comprising: measuring a fourth positioning performance of the specimen positioning system during the predetermined movement at each of one or more locations in the workspace of the specimen positioning system based on the updated set of values corresponding to the one or more feedforward control parameters.

**8.** A semiconductor measurement system, comprising: a specimen positioning system comprising: at least one motion stage configured to move a specimen within a range of locations in at least one degree of freedom, the range of locations comprising a workspace of the specimen positioning system; a measurement subsystem including one or more sensors configured to measure a motion of the specimen; a motion controller configured to control the motion of the specimen, the motion controller including at least one feedforward control element characterized by one or more feedforward control parameters; and a computing system configured to: measure a first positioning performance of the specimen positioning system during a predetermined movement at each of one or more locations in the workspace of the specimen positioning system, the motion controller controlling the predetermined movement at each of the one or more locations in the workspace, the predetermined movement undertaken at a first set of values corresponding to the one or more feedforward control parameters; simulate a second positioning performance of the specimen positioning system during the predetermined movement at each of one or more locations in the workspace based on the first set of values corresponding to the one or more feedforward control parameters; simulate a third positioning performance of the specimen positioning system during the predetermined movement at each of one or more locations in the workspace based on an updated set of values corresponding to the one or more feedforward control parameters; and select the updated set of values corresponding to the one or more feedforward control parameters to minimize a cost function including the second positioning performance subtracted from a sum of the first and third positioning performances.

**9.** The semiconductor measurement system of claim 8, wherein the first, second and third positioning performances are positioning errors after a predetermined time after initiation of the predetermined movement.

**10.** The semiconductor measurement system of claim 8, wherein the first, second, and third positioning performances are positioning errors after a predetermined time after termination of the predetermined movement.

**11.** The semiconductor measurement system of claim 8, wherein the one or more locations in the workspace includes multiple, different locations in the workspace in a direction aligned with a first

degree of freedom of the specimen positioning system.

**12.** The semiconductor measurement system of claim 11, wherein the one or more locations in the workspace includes multiple, different locations in the workspace in a direction perpendicular with the first degree of freedom.

**13.** The semiconductor measurement system of claim 8, wherein the workspace of the specimen positioning system includes at least two degrees of freedom.

**14.** The semiconductor measurement system of claim 8, further comprising: measuring a fourth positioning performance of the specimen positioning system during the predetermined movement at each of one or more locations in the workspace of the specimen positioning system based on the updated set of values corresponding to the one or more feedforward control parameters.

**15.** A semiconductor measurement system, comprising: a specimen positioning system comprising: at least one motion stage configured to move a specimen within a range of locations in at least one degree of freedom, the range of locations comprising a workspace of the specimen positioning system; a measurement subsystem including one or more sensors configured to measure a motion of the specimen; a motion controller configured to control the motion of the specimen, the motion controller including at least one feedforward control element characterized by one or more feedforward control parameters; and a non-transitory, computer-readable medium storing instructions that, when executed by one or more processors, causes the one or more processors to: measure a first positioning performance of the specimen positioning system during a predetermined movement at each of one or more locations in the workspace of the specimen positioning system, the motion controller controlling the predetermined movement at each of the one or more locations in the workspace, the predetermined movement undertaken at a first set of values corresponding to the one or more feedforward control parameters; simulate a second positioning performance of the specimen positioning system during the predetermined movement at each of one or more locations in the workspace based on the first set of values corresponding to the one or more feedforward control parameters; simulate a third positioning performance of the specimen positioning system during the predetermined movement at each of one or more locations in the workspace based on an updated set of values corresponding to the one or more feedforward control parameters; and select the updated set of values corresponding to the one or more feedforward control parameters to minimize a cost function including the second positioning performance subtracted from a sum of the first and third positioning performances.

**16.** The semiconductor measurement system of claim 15, wherein the first, second and third positioning performances are positioning errors after a predetermined time after initiation of the predetermined movement.

**17.** The semiconductor measurement system of claim 15, wherein the first, second, and third positioning performances are positioning errors after a predetermined time after termination of the predetermined movement.

**18.** The semiconductor measurement system of claim 15, wherein the one or more locations in the workspace includes multiple, different locations in the workspace in a direction aligned with a first degree of freedom of the specimen positioning system.

**19.** The semiconductor measurement system of claim 18, wherein the one or more locations in the workspace includes multiple, different locations in the workspace in a direction perpendicular with the first degree of freedom.

**20.** The semiconductor measurement system of claim 15, wherein the workspace of the specimen positioning system includes at least two degrees of freedom.

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